

Fig. 42.1 Depiction of the relative contributions of cardiovascular disease types across chronic kidney disease stages. Atherosclerotic and nonatherosclerotic disease may contribute synergistically to increased cardiovascular risk as kidney function declines. CV, Cardiovascular; GFR, glomerular filtration rate. (Adapted from Wanner C, Amann K, Shoji T. The heart and vascular system in dialysis. *Lancet*. 2016;388:276–284.)

Fig_42_1

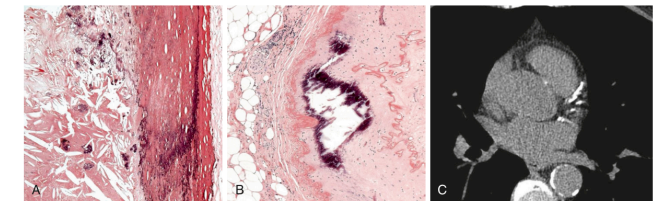


Fig. 42.2 Arterial calcification in chronic kidney disease (CKD). Cross-sections of medium-sized arteries from a patient with CKD showing deposition of calcium (black) in the intima (A) and media (B) in association with atherosclerosis (von Kossa stain). Calcium deposits may be visible on computed tomography scanning of the heart as depicted in (C) where calcification is visible in the left anterior descending and left circumflex coronary artery, as well as the descending aorta. (Courtesy Professor A.J. Howie.)

Fig_42_2

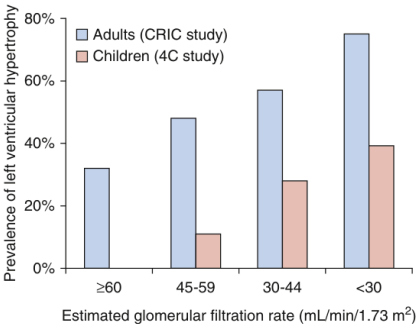


Fig. 42.3 Prevalence of left ventricular hypertrophy in adults and children with chronic kidney disease.

Fig_42_3



Fig. 42.4 Left ventricular disease in chronic kidney disease (CKD). A cross-section of a postmortem heart from a patient with longstanding CKD showing concentric left ventricular hypertrophy (A). Histologic analysis (B) often reveals myocardial fibrosis (pale staining) disrupting the normal architecture of cardiac myocytes. (Courtesy Professor A.J. Howie [A] and Dr. M. Rubens [B].)

Fig_42_4

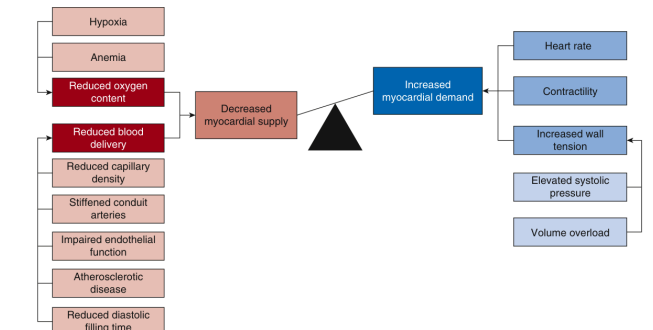


Fig. 42.5 Pathogenesis of myocardial fibrosis as a function of oxygen supply-demand mismatch. Multiple factors result in decreased supply (reduced oxygen content and blood delivery) and increased demand (increased heart rate, contractility, and wall tension).

Fig_42_5

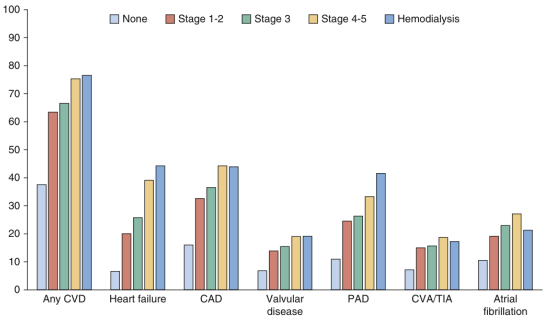


Fig. 42.6 Age-, sex-, and race-adjusted prevalence of cardiovascular disease (CVD) and its subtypes in individuals with chronic kidney disease. CAD, Coronary artery disease; CVA, cerebrovascular accident; PAD, peripheral artery disease; TIA, transient ischemic attack. (Data are derived from the USRDS 2020 Annual Data Report, Volume 1, Chapter 4 and Volume 2, Chapter 8. Available at: <https://adr.usrds.org/2020/chronic-kidney-disease/4-cardiovascular-disease-in-patients-with-ckd>. Accessed September 29, 2023.)

Fig_42_6

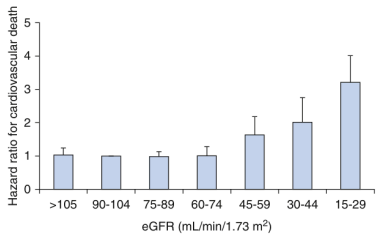


Fig. 42.7 Hazard ratios for cardiovascular events according to the baseline estimated glomerular filtration rate. Adjusted for age, sex, race, cardiovascular disease history, smoking status, diabetes mellitus, systolic blood pressure, serum total cholesterol, and urine albumin-to-creatinine ratio. (Plotted with data from van der Velde M, Matsushita K, Coresh J, et al. Lower estimated glomerular filtration rate and higher albuminuria are associated with all-cause and cardiovascular mortality. A collaborative meta-analysis of high-risk population cohorts. *Kidney Int*. 2011;79:1341–1352.)

Fig_42_7

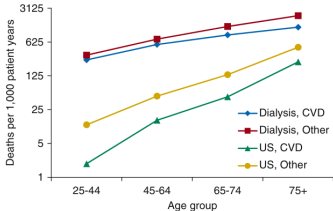


Fig. 42.8 Cardiovascular disease (CVD) and noncardiovascular disease mortality in the general U.S. population (2014) compared with patients with chronic kidney failure treated by dialysis (2012–2014). Cardiovascular disease (CVD) deaths in the U.S. population include “Diseases of the heart” (100–109, 111, 113, 120–151) and “Cerebrovascular diseases” (160–169). CVD death in dialysis includes myocardial infarction, pericarditis, atherosclerotic heart disease, cardiomyopathy, arrhythmia, cardiac arrest, valvular heart disease, congestive heart failure, and cerebrovascular disease. The youngest general population group is 22 to 44 years old. (Reprinted from Gilbert SJ, Weiner DE, Bombardier AS, et al. *The Primer on Kidney Diseases*. 8th ed. Philadelphia: Elsevier; 2022.)

Fig_42_8

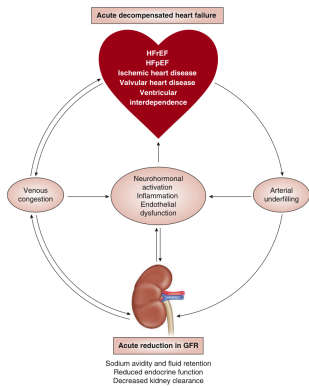


Fig. 42.9 Proposed pathways leading to the cardinal syndrome and its complications. The inciting event is usually an acute decompensation of heart failure, resulting in venous congestion and/or arterial underfilling that promotes neurohormonal activation, inflammation, and endothelial dysfunction. Clinically, this manifests with reductions in GFR, as well as increasing sodium avidity and fluid retention, all of which further perpetuate the process. (Reprinted from McKelvie W, Sarnak MJ. Cardiorespiratory syndrome in the hospital. *CMAJ*. 2003;169:933–945.)

Fig_42_9

Table 42.1 Manifestations of Cardiovascular Diseases in Persons With Chronic Kidney Disease

Arteries	Heart
Increased wall thickness	Altered cardiac geometry
Arterial stiffness	Myocardial fibrosis
Endothelial dysfunction	Left ventricular dysfunction
Arterial calcification	Valvular disease
Atherosclerosis	Dysrhythmia and conduction defects

Table_42_1

Table 42.2 Traditional and Nontraditional Cardiac Risk Factors in Chronic Kidney Disease

Traditional Risk Factors	Nontraditional Factors
Older age	Albuminuria
Male sex	Lipoprotein (a) and apo (a) isoforms
Hypertension	Lipoprotein remnants
Higher LDL cholesterol	Anemia
Lower HDL cholesterol	Abnormal mineral metabolism
Diabetes	Extracellular fluid volume overload
Smoking	Electrolyte abnormalities
Physical inactivity	Oxidative stress
Menopause	Inflammation
Family history of cardiovascular disease	Malnutrition
Left ventricular hypertrophy	Thrombogenic factors
	Sleep disturbances
	Altered nitric oxide/endothelin balance
	Sympathetic overactivity

HDL, High-density lipoprotein; LDL, low-density lipoprotein. From Sarnak MJ, Levey AS. Cardiovascular disease and chronic renal disease: a new paradigm. *Am J Kidney Dis.* 2000;35(4 suppl 1):S117–S131.

Table_42_2

Table 42.3 Imaging Modalities for Coronary Disease

	Stress Echocardiography	Gated SPECT Myocardial Perfusion Imaging	Gated PET Myocardial Perfusion Imaging	Coronary CT Angiography	Pharmacologic Stress Cardiac MR	Left Heart Catheterization
Strengths	-Assesses heart structure and function -Identifies regional wall abnormalities -Good sensitivity and specificity -Provides information on myocardial blood flow	-Widely available -Good sensitivity and specificity -Provides information on LV function	-Highly accurate -Provides information on myocardial blood flow	-Aligns with prognosis	-Combined assessment of LV function, myocardial perfusion, scarring and wall motion -Does not require the geometric assumptions used in 2D echo	-Allows for concurrent diagnosis and evaluation -Assessment of flow limiting lesions
Limitations	-Poor acoustic windows may limit accuracy -Inadequate stress may result in a false negative -User expertise	-Soft tissue attenuation and artifacts may decrease test performance	-Expensive -Limited availability	-Limited specificity in individuals with calcification, common in advanced CKD	-Expensive -Limited availability -Device compatibility	-Invasive -Expensive
Safety in CKD	No issues	No issues	No issues	Iodinated contrast used	Gadolinium used	Iodinated contrast used; arterial access required

LV, Left ventricle; MRI, myocardial perfusion imaging. Adapted from Diszian V, Gewirtz H, Marwick TH, et al. Cardiac imaging for coronary heart disease risk stratification in chronic kidney disease. *JACC Cardiovasc Imaging.* 2021;14:669–682.

Table_42_3

Table 42.4 Randomized Controlled Trials of Statin Treatment Specifically in Chronic Kidney Disease

Study	Intervention	Population	Median Follow-Up	Primary Outcome	Risk of Primary Outcome	Risk of All-Cause Mortality
4D ⁴³	Atorvastatin 20mg daily (vs. placebo)	Type 2 diabetes, HD <2 years, LDL 80–180mg/dL	4.0 years	Composite of death from cardiac causes, fatal stroke, nonfatal MI, or nonfatal stroke	RR = 0.92 (0.77–1.10)	RR = 0.93 (0.79–1.08)
AURORA ⁴⁴	Rosuvastatin 10mg daily (vs. placebo)	HD or HDF >3 months	3.8 years	Composite of death from cardiovascular causes, nonfatal MI, or nonfatal stroke	RR = 0.96 (0.84–1.11)	RR = 0.96 (0.86–1.07)
ALERT ⁴⁵	Fluvastatin 40mg daily (vs. placebo)	>6 months post kidney transplant with stable graft function; no recent MI; total cholesterol 155–348mg/dL	5.4 years	Major adverse cardiac event defined as cardiac death, nonfatal MI, or coronary revascularization procedure	RR = 0.83 (0.64–1.06)	RR = 1.02 (0.81–1.30)
SHARP ⁴⁶	Simvastatin 20mg daily + ezetimibe 10mg daily (vs. placebo)	No previous MI or coronary revascularization; Creatinine >1.7mg/dL (men) or >1.5mg/dL (women)	4.9 years	Composite of coronary death, nonfatal MI, ischemic stroke, or any revascularization procedure	RR = 0.83 (0.74–0.94)	RR = 1.02 (0.94–1.11)

Data in parentheses represents 95% confidence intervals. RR and RI report the relationship between treatment versus placebo, with values below 1 favoring treatment and above 1 favoring placebo. In SHARP there was no statistically significant difference in the risk of the primary outcome between dialysis and nondialysis patients ($P = 0.25$), although, in subgroup analyses, there was no benefit in participants receiving HD at randomization (RR = 0.98 [0.78, 1.19], 4D. German Diabetes Dialysis Study. *ALERT*, assessment of Lacosin in renal transplantation; AURORA, a study to evaluate the use of rosuvastatin in subjects in regular hemodialysis; an assessment of survival and cardiovascular events; HD, hemodialysis; HDF, hemodiafiltration; RR, hazard ratio; LDL, low-density lipoprotein; MI, myocardial infarction; RR, risk ratio; SHARP, Study of Heart and Renal Protection. Adapted from National Kidney Foundation. *Primer on Kidney Diseases*. 8th ed. Philadelphia: Elsevier; 2022.

Table_42_4